Same-u mixed UInt64 correspondence
Original-input branch composition

Result
Three declarations compiled with separate axiom readback: cross_word_decode, same_point_decode and same_u_word_preserves. The first two establish literal UInt64 expression canonicality and field decoding. The third composes original-input same-u field validity with ordered word guards and the existing literal doubling bridge.

Premises
Prime5<=p<2^31; twelve canonical input coordinate words and four canonical curve coefficient words; four nonzero representation weights; both original normalized validTwo premises; literal crossed support equalities and earlier inverse/equal guard exclusions; global IsCoprime f f.derivative. Field.n=1/tables=None is the source arithmetic interpretation. Operational counter headroom is outside this pure value model.

Literal source mapping
Cross-coordinate ASTs preserve square/cube/fourth/fifth-power multiplication association. Extracted k,x,y,z,w preserve the retained gate903-958 tree. There is no source pow call in this extraction path. FieldExpr composition supplies canonicality and arithmetic no-overflow correspondence for products/sums under the prime bound; the existing doubling bridge handles raw literal5 including p=5.

Guard and validity composition
Canonical equality and wordNeg reflection transport original crossed guards to reviewed SameUComposition. This yields nonzero sourceK and extracted ordinate from original validity/global coprimality. Literal doubled payload decoding transports validTwo and nonzero output weights back to canonical output words. The scoped dispatcher tests support, inverse, equality and zero slope in native order; None on earlier/other branches is a restriction marker, not the corresponding production payload.

Fixed controls
A full theorem application uses nonunit original words a=(0,4,0,1,2,3), b=(0,4,4,0,3,2), p=5, curve coefficients(1,4,0,0). It reuses exact original-input polynomial witnesses and explicit global certificate.
The extracted k=4, point=(1,2,4,3), and doubled raw payload=(3,1,2,4,4,4) are independently asserted by fixed word evaluation. Equal/inverse restricted routes return None; noncanonical coordinate5 at p5 is rejected as a theorem premise.

Evidence and failures
Final main attempt3, controls2 and readback1 exited0. Initial main parse failure used reserved identifier at; first controls lacked an explicit UInt64-prime instance and structural equality/canonical simplification. All attempts remain preserved. The three readbacks list only propext, Classical.choice and Quot.sound. The exact copied field-fixture module is separately compiled and bound.

Scope
Independent source-fidelity review pending. No divisor class-sum, whole degree2+2 dispatcher, compiled implementation, counter correctness or performance claim is made. No native scientific trial or shared-source/cache edit occurred. Cost ratio is null.